Arthrex ACP™ Double Syringe System

Autologous **C**onditioned **P**lasma

Biologics in Orthopedics

- Biologics has a solid history in Europe and some use in North America
  - Now gaining in popularity for orthopedic applications
  - Coining of the phrase “Regenerative Medicine”
- Why?
  - Increasing older population wants to maintain active lifestyle
  - We treat areas with limited blood supply and slow cell turnover
  - Greater demands on surgeons to produce more rapid return to sport

A Practical Approach

- Serious challenges for use of biologics in orthopedic applications
  - A system must:
    - Be affordable
    - Cannot add significant time to the procedure
    - Be easy to use
  - Corporate and Scientific Challenges
    - Regulatory control
    - Cost and time for development
    - Quantifying results

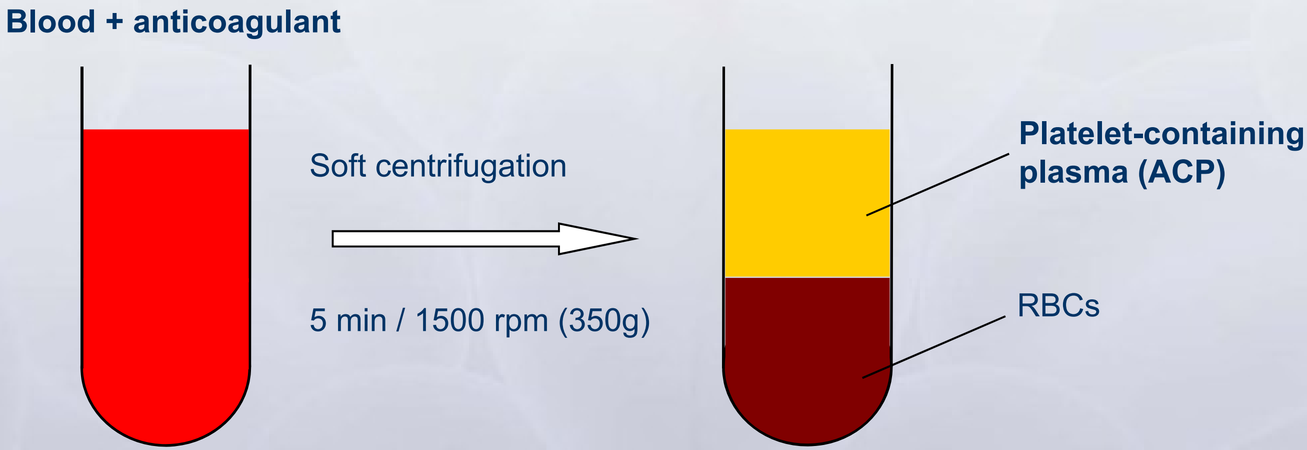
Our Answer - ACP

- **A**utologous **C**onditioned **P**lasma
- Platelet concentrate for use in orthopaedic applications
- Increased concentration of platelets and growth factors which are associated with the healing process

What Exactly Is ACP?

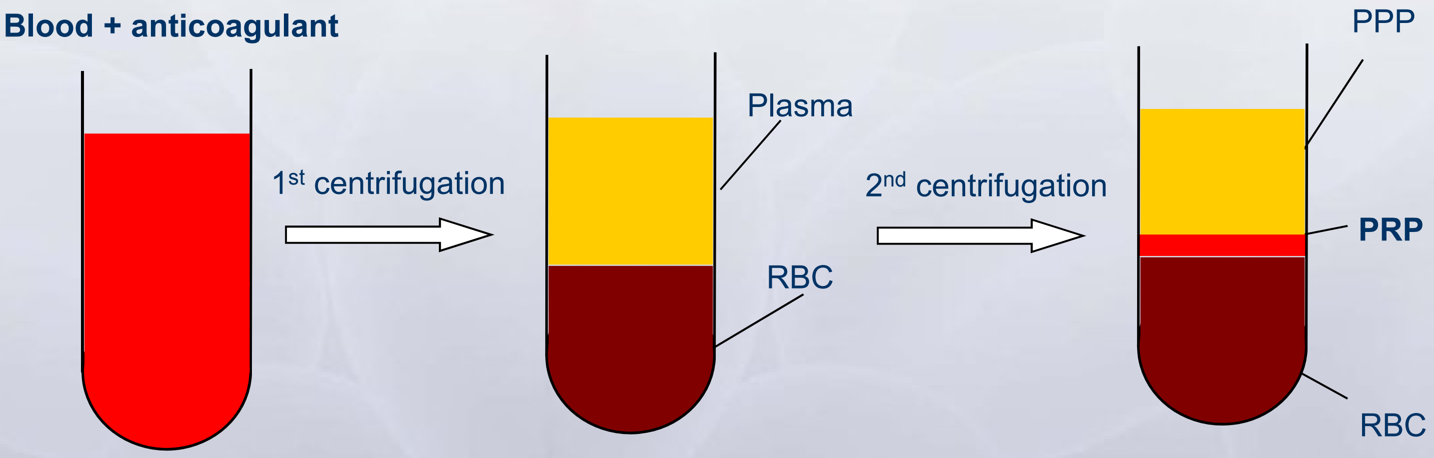
- An easy to use system that concentrates platelets and growth factors within a plasma layer **separate** from red and white blood cells
- Growth factors and other molecules within the plasma layer may modulate healing

What Does ACP Look Like?



- Easy system to use
- Minimal additional capital equipment
- Additional procedure time of ~5-10 minutes

What Do Other Systems Look Like?



- Complicated systems that require rep support
- Expensive capital and expensive kits
- Additional procedure time (15 - 30 minutes)

FDA Clearance

DEPARTMENT OF HEALTH & HUMAN SERVICES

[illegible]

Regulatory Affairs Associate

Naples, FL [illegible]

Re: [illegible]

Device Name: [illegible] Double Syringe (ACP) [illegible]

Regulation Number: 21 CFR [illegible]

Regulatory Class: Class II

Received: September [illegible], 2008

[illegible]

1 Indications for Use Form

Indications for Use

510(k) Number: [illegible]

Device Name: *Arthrex Double Syringe (ACP) System*

The *Double Syringe (ACP) System* is used to facilitate the safe and rapid preparation of autologous platelet-rich-plasma (PRP) from a small sample of blood at the patient's point of care.

The PRP can be mixed with autograft and allograft bone prior to application to an orthopedic surgical site as deemed necessary by the clinical use requirements.

Prescription Use ✓ AND/OR Over-The-Counter Use ____
(Per 21 CFR 801 Subpart D) (21 CFR 801 Subpart C)

(PLEASE DO NOT WRITE BELOW THIS LINE-CONTINUE ON ANOTHER PAGE IF NEEDED)

Concurrence of CBER, Office of Cellular, Tissue and Gene Therapies (OCTGT)

Director, OCTGT

[illegible]

510(k) Number [illegible]

Using ACP in Medical Practice

Syringe, ACD-A, and cap

Withdraw blood + mix with ACD-A

Centrifuge 1500 rpm for 5 min

Transfer plasma into smaller syringe and unscrew

ACP ready to use

Using ACP in the OR

Syringe, ACD-A, cap, and sterile bucket

Scrub nurse withdraws blood + mixes with ACD-A

Scrub nurse places syringe into bucket and passes closed bucket to circulating nurse

Centrifuge 1500 rpm for 5 min

Circulating nurse opens cap and scrub nurse removes sterile syringe

Transfer plasma into smaller syringe and unscrew

ACP is transferred into sterile basin on sterile field. A sterile 5 cc syringe is used to recover ACP and use at the point of care.

ACP platelet concentration

Higher vs. Lower Concentration

- PRP concentration is patient-specific
  - Pietrzak and Eppley, 2005; Weibrich *et al,* 2001; Anitua *et al*, 2004
    - Dependent on injury, patient condition, processing variables (single vs. double spin, PRP volume, etc.)
    - **Not necessary to have high concentration for effective healing**
    - **Higher GF concentrations can be detrimental to healing**

Higher vs. Lower Concentration

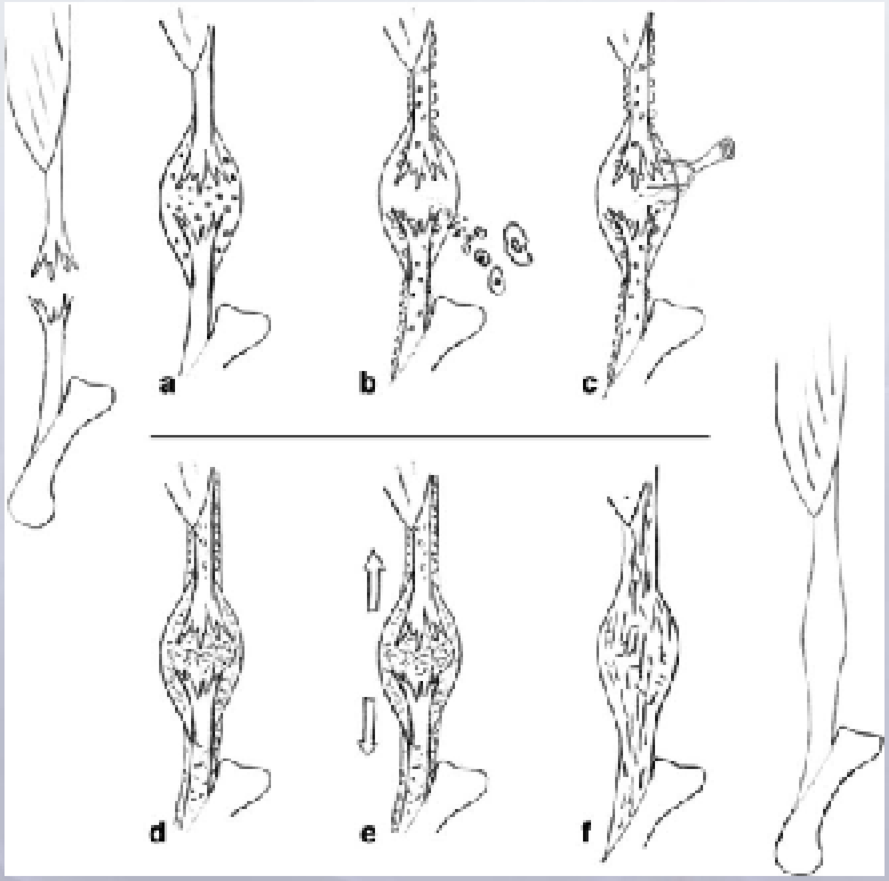
- Anitua's definition of platelet concentration
  - Whole blood baseline – 1.5 - 2 x 10^5 platelets / µL
  - **Minimum 3 x 10^5 platelets / µL – 2X PRP**
- **2 - 3X ACP concentration modulates healing**
  - Other systems use similar concentrations
    - Plasma rich in growth factors (PRGF) – Spain
    - Clinaseal centrifuge - US
    - Nahita centrifuge - Spain

Higher vs. Lower Concentration

- Marx's definition of PRP concentration
  - Whole blood baseline – 1.5 - 2 x 10^5 platelets / µL
  - **Minimum 1 x 10^6 platelets / µL - 5X PRP**

The science of ACP

Healing Cascade



- a – Hematoma with platelet activation
- b – Invasion of cells and proliferation of paratenon
- c – Vascular and neuronal ingrowth
- d – Loose collagenous callus formation
- e – Mechanical stimulation
- f – Maturation and remodeling

Taken from Aspenberg, Int Orthop 2007; 31(6): 783-9

Most systems concentrate RBCs and WBCs – why this is bad

- Scott A *et al*. What do we mean by the term "inflammation"? A contemporary basic science update for sports medicine. Br J Sports Med 2004; 38(3): 372-80.
  - **Activated WBCs release MMPs that degrade matrix and reactive oxygen species that destroy anything close by - can lead to inflammation**
- Jiang N *et al*. Respiratory protein–generated reactive oxygen species as an antimicrobial strategy. Nat Immunol 2007; 8(10): 1114-22.
  - **Lysed RBCs release free radicals, similar to WBCs, that destroy anything close by – good for bacteria, not good for platelets**

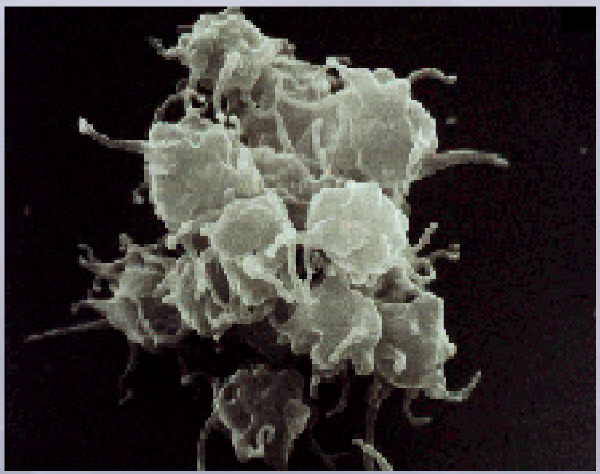
Why is this resultant PRP RED?

What Are Platelets?

- Fragments of cells derived from megakaryocytes in bone marrow that live between 5-9 days
  - They do not contain a nucleus
  - Contain growth factors and other molecules to modulate healing and hemostasis
  - Platelets release growth factor-containing α-granules into plasma

Platelet Activation

Unactivated platelets



Activated platelets

- Releases growth factors and other cytokines from α-granules
  - Calcium chloride or any other calcium salt
  - Thrombin
- Centrifugation activates platelets, but to much smaller extent

ACD-A

- All PRP systems require the addition of an anticoagulant (ACD-A)
  - These additives drive down the pH and dilute the end product
  - Why is this a problem?
  - Possible solution to add sodium bicarbonate
- When to use or not to use?
  - <30 minutes - no need
  - >30 minutes - use ACD-A, will stay for 4 hours

Growth Factors

- Promote collagen formation
- Stimulate angiogenesis
- Modulate matrix formation

Growth Factors in ACP

Growth Factor	Phase in Which Most Active	Functions
IGF-1	Inflammation, proliferation	Promotes proliferation and migration of cells, stimulates matrix production
TGF-β	Inflammation	Regulates cell migration, proteinase expression, fibronectin binding interactions, termination of cell proliferation, stimulation of collagen production
VEGF	Proliferation, remodeling	Promotes angiogenesis
PDGF	Proliferation, remodeling	Regulates protein and DNA synthesis at injury site, regulates expression of other growth factors
bFGF	Proliferation, remodeling	Promotes cellular migration, angiogenesis
EGF	Proliferation, remodeling	Stimulates proliferation and differentiation of epidermal cells, stimulates angiogenesis

Adapted from Molloy *et al*, Sports Med 2003; 33(5): 381-94

How does ACP compare?

Comparison of ACP vs. PRP

	Arthrex ACP	Other PRP Systems
Volume drawn	10 mL	60-120 mL
Is anticoagulant (ACD-A) necessary?	No: if ACP is used within 30 minutes	Yes
Centrifugation steps	1X	1-2X
Centrifugation time	5 min	15-30 min
Does it concentrate red and white blood cells?	No: < whole blood levels	Yes: >= whole blood levels
Cost	$150	$350-900
Harvest volume	2-5 mL	6-10 mL
Growth Factor Concentration	~5X -25X	~5-10X
Mode of action	Growth factors released by platelets, mixed with proteins and electrolytes in plasma	Growth factors released by platelets
Platelet concentration/µL	300,000 – 500,000	500,000 – 1,000,000

Comparison to Other Systems

Device Name	IGF-1 Increase	TGF-β Increase	VEGF Increase	PDGF-AB Increase	PDGF-BB Increase	EGF Increase
Arthrex ACP	***1X***	***4X***	***11X***	***25X***	***6X***	***5X***
[1]Biomet GPS™	1X	3.6X	6.2X	N/A	5.1X	3.9X
[2]Harvest® SmartPrep2™	N/A	4.4X	4.4X	4.4X	N/A	4.4X
[3]DePuy Symphony II	N/A	3-6X	2-3X	4-5X	N/A	4-6X
[4]Medtronic Magellan™	N/A	4-6X	3-6X	6-10X	N/A	8-10X
[5]Cascade Fibrinet	5-10X*	5-10X*	5-10X*	N/A	5-10X*	5-10X*

1. GPS – From Eppley *et al*, Plast Reconstr Surg, 114: 1502, 2004 and Biomet website
2. SmartPrep2 – From Harvest Technologies brochure
3. Symphony II – From DePuy White Paper
4. Magellan – From Medtronic brochure
5. Fibrinet – From Cascade Medical brochure, presentation, and White Papers

* Fibrinet growth factor levels monitored over a 7 day period - all others were baseline

Our Competition's Story

- What are they going to say?
  - Lower platelet concentration levels
  - The product from ACP is PPP
    - Not true!
  - Delivery method: To clot or not?
    - Using thrombin = immediate release of GF
    - Kit design (Micromedics' FibriJet parts)
  - Where are the studies?

Key Features and Benefits

- Affordable
- Easy to use and short procedure time
- Requires less blood than conventional PRP systems
- Anticoagulant not necessary in all cases
  - Less dilution
  - More natural pH
- Reduced level of RBC and WBCs vs. baseline
- Similar GF concentration to other systems

PRP Gelling Agent Solution

- Gelling agent powder – 4800-5000 IUs (International Units)
- Gelling agent liquid – 5 mL of 10% solution
- Mix 1000 IUs of gelling agent powder with 1 mL of gelling agent liquid (1000 IUs/mL ratio of powder/liquid)
  - 5000 IUs of powder to 5 mL of liquid
  - 10000 IUs of powder to 10 mL of liquid
  - This creates the gelling agent solution
- Mix 1 mL of PRP with 0.1 mL of gelling agent solution to create a clot/gel (10:1 ratio of PRP/gelling agent solution)
  - Can be prepared with syringe mixer system or in sterile cup
  - If clot prepared in sterile cup, transfer clot to dry sterile cup after forming to make it more solid

Capital Program

- Two Program Options
  - If the capital equipment is placed and disposables $200
    - Minimum purchase $8,000 in disposables over 12 months
    - Disposables convert to $150 after minimum purchase is met
  - If the capital equipment is placed and disposables $150
    - Minimum purchase $8,000 in disposables over 12 months
    - Reduced commissions until minimum purchase is met
  - Capital Equipment Cost $7K
  - Distributorship may purchase capital under conventional Arthrex capital agreement

Coding / Reimbursement

ACP Reimbursement Coding

- Arthrex recommends that you check with your insurance provider or payer for reimbursement

ACP Procedure Pearls

ACP Pearls

- Prior to the case
  - Prime the syringe
  - Make sure the inner syringe is screwed in tightly
- Phlebotomy
  - Blood draw should be performed with a 18-20G butterfly needle
  - Draw blood **SLOWLY** (1cc / 2 seconds)
- Separate ACP as soon as possible after blood draw
- If using within 30 min of blood draw, do not use ACD-A
- ACP with ACD-A can stay up to 4 hours at room temperature before use, but use is recommended within 2 hours

CENTRIFUGE

UNPACKING & SETUP

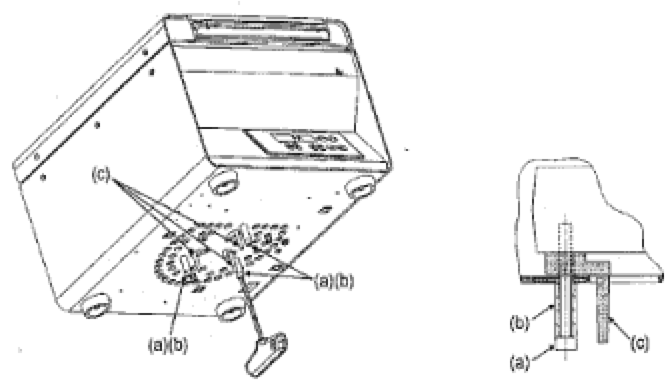
EVERY CENTRIFUGE SHOULD HAVE A SHIPPING INSERT LIKE THIS PACKED WITH IT, USUALLY INSIDE THE CENTRIFUGE CHAMBER

ROTOFIX 32A

ACHTUNG
Transportsicherung unbedingt entfernen

CAUTION
Remove transport protection unconditionally

ATTENTION
Retirer imperativement la protection de transport



1. Die 3 Schrauben (a) und Abstandshülsen (b) entfernen.
2. Die Transportsicherung (c) 3x entfernen.

1. Remove the 3 screws (a) and distance sleeves (b).
2. Remove the transport protection (c) 3x.

1. Enlever les 3 vis (a) et ses capuchons (b).
2. Enlever les protections de transport (c) 3x.

Achtung:	Die Transportsicherung muss vor jedem Transport eingebaut werden.
Caution:	The transport protection has to be installed before each transportation.
Attention:	La protection de transport doit être installée avant chaque transport.

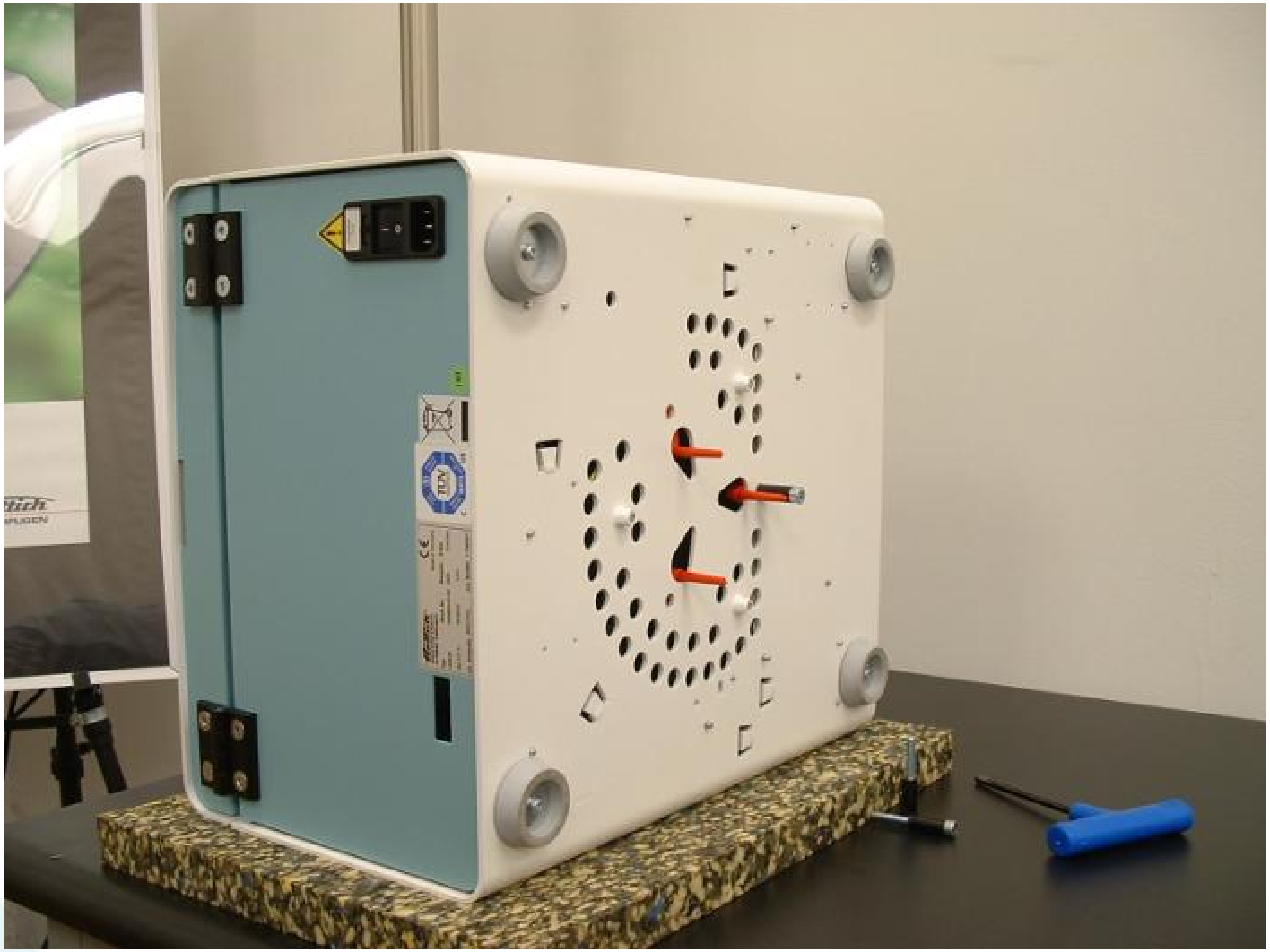
INCLUDED WITH ROTOFIX 32A

- *SHORT T-HANDLE (E613)*
- *GREASE FOR ROTOR (4051)*
- *13 AMP POWER CABLE (6083)*
- *PLASTIC EMERGANCY OPEN PIN (E2914)*
- *TWO SPARE FUSES (E914)*
- *OPERSTOR’S MANUAL*
- *SHIPPING INSERT*

***** Numbers in ( ) are part numbers***

Hettich

UNPACKING & SETUP

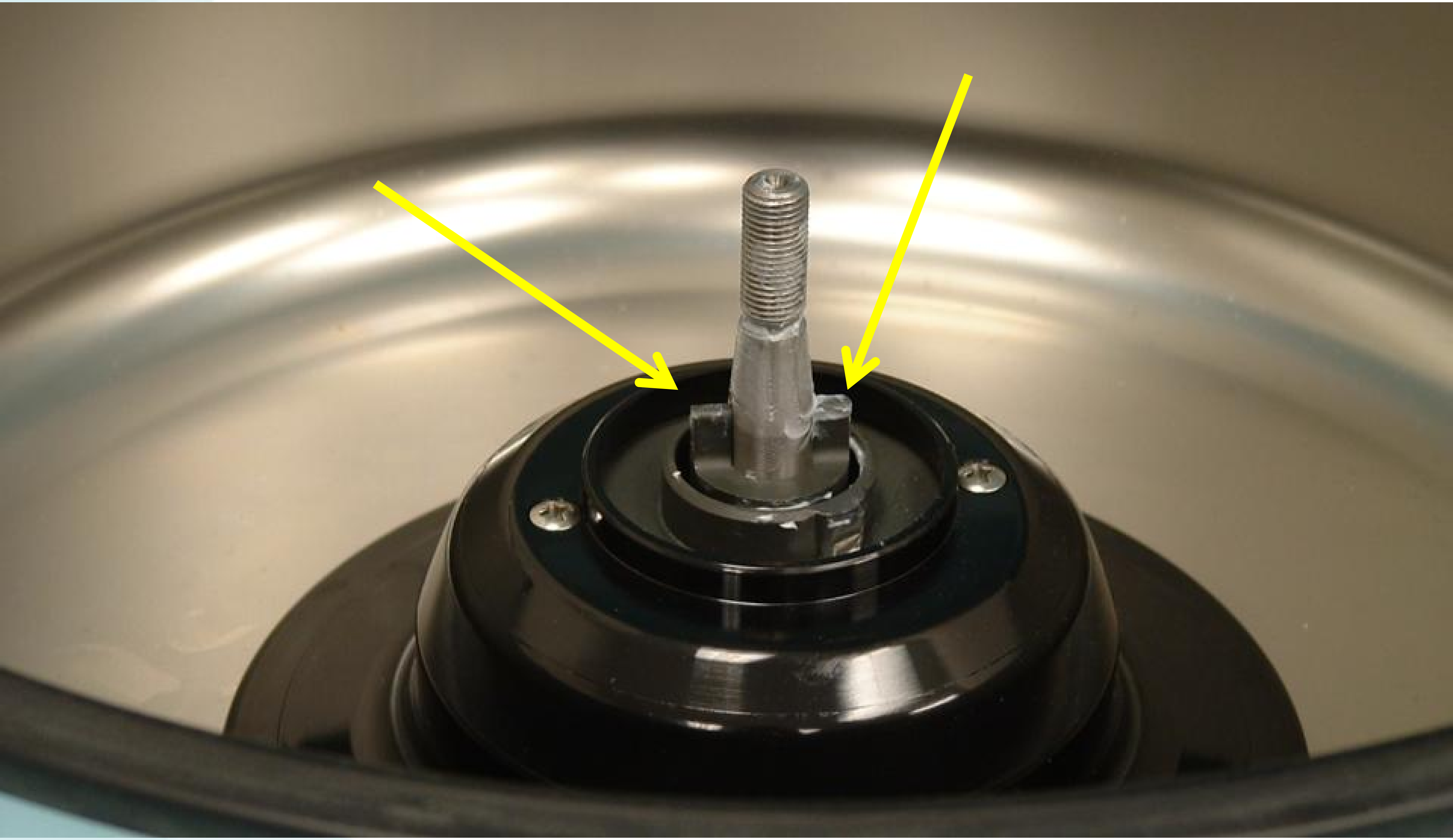


UNPACKING & SETUP

WHEN INSTALLING THE ROTOR, APPLY A THIN COATING OF HETTICH GREASE ON THE MOTOR SHAFT TO PREVENT ROTOR FROM BINDING TO THE SHAFT OVER TIME.

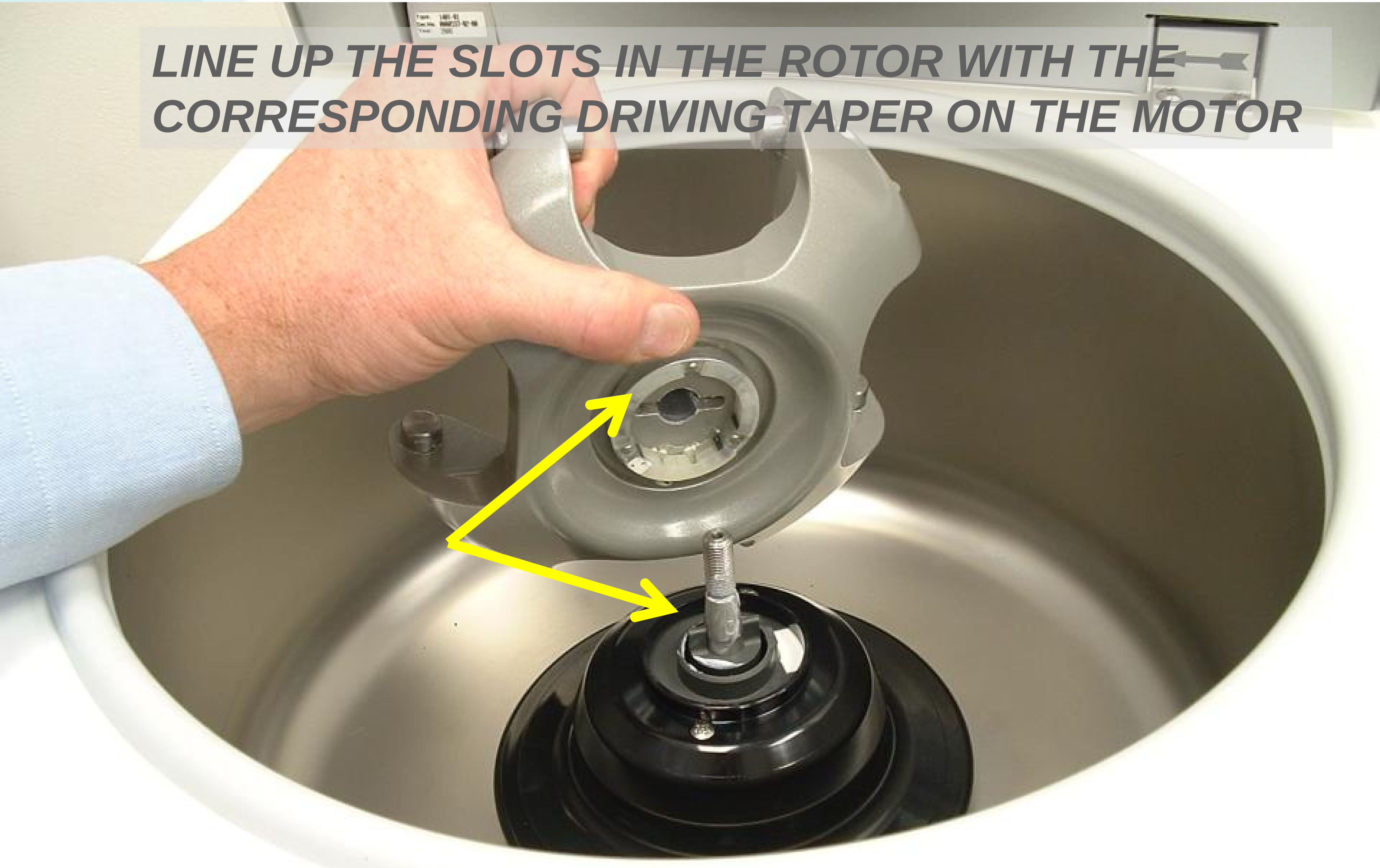
PART # 4051


Hettich
ZENTRIFUGEN
Schmierstoff für die
Tragzapfen am Rotor
Anwendung: 1 mal wöchentlich
Schmierstoff, zugelassen vom USDA. Gruppe H 1
Bestell-Nr. 4051 / Entsorgung: gem. den gesetzl. Auflagen
Mindesthaltbarkeitsdatum: Juli 2010
Grease for the trunnions of the rotor. Lubricant,
approved by the USDA, group H 1 / order no. 4051
Application: once a week

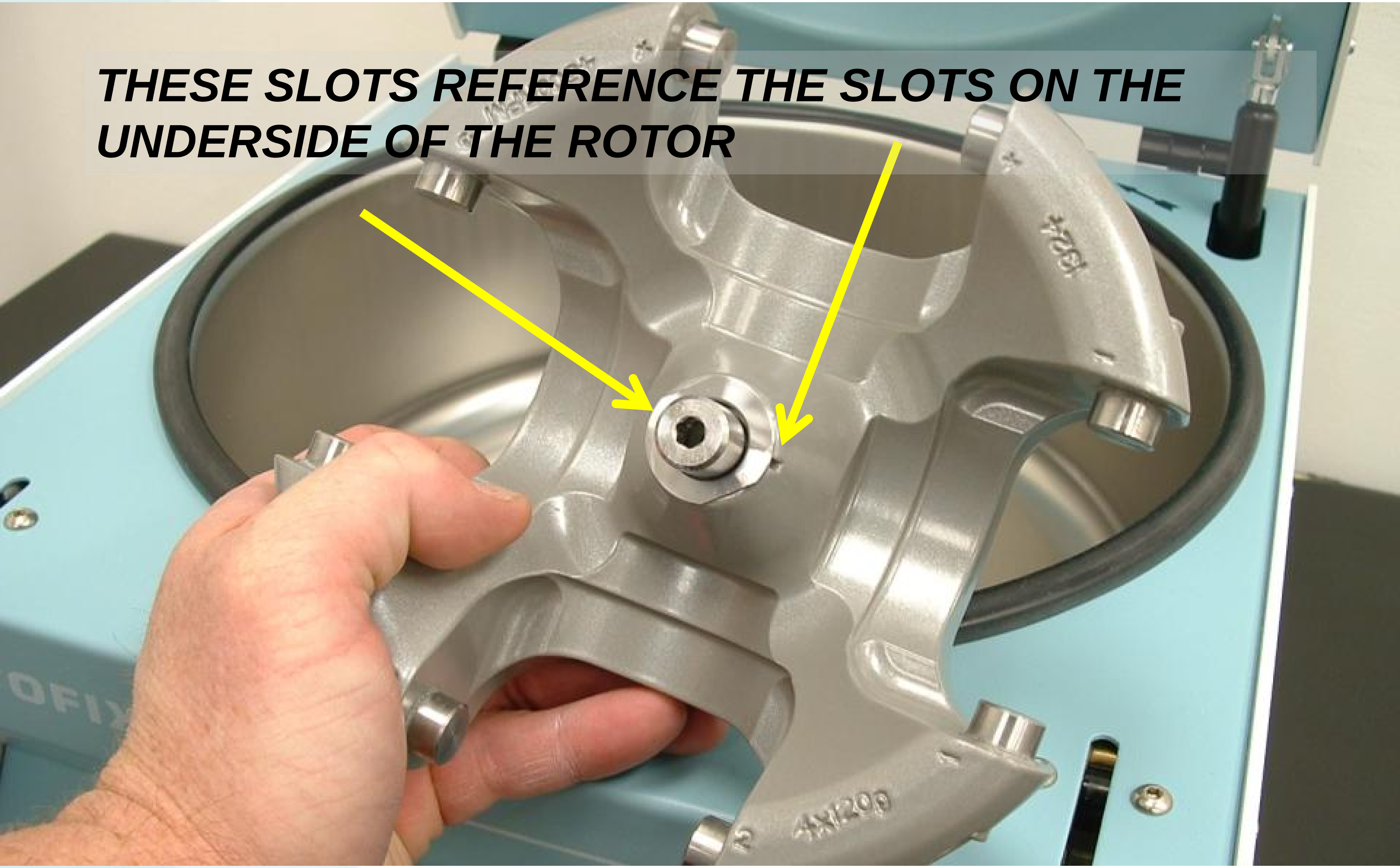


UNPACKING & SETUP

LINE UP THE SLOTS IN THE ROTOR WITH THE CORRESPONDING DRIVING TAPER ON THE MOTOR



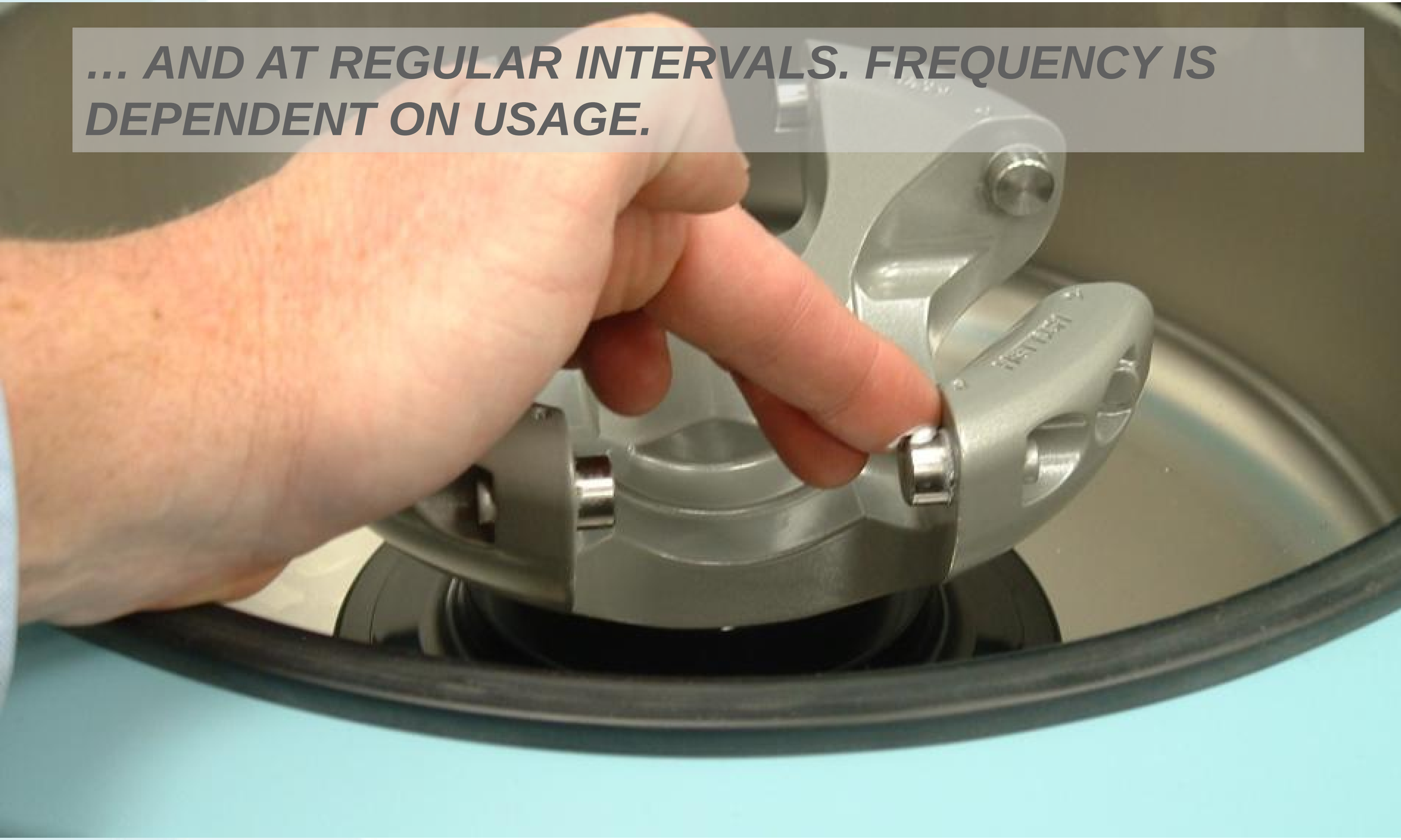
THESE SLOTS REFERENCE THE SLOTS ON THE UNDERSIDE OF THE ROTOR



UNPACKING & SETUP

GREASE EACH TRUNNION AT INITIAL SET-UP…

… AND AT REGULAR INTERVALS. FREQUENCY IS DEPENDENT ON USAGE.



UNPACKING & SETUP

Bucket installation

- MAKE SURE TRUNNIONS ARE GREASED
- ALL FOUR BUCKETS MUST BE INSTALLED

PROGRAMMING THE ROTOFIX 32A

- ***USE THE UP AND DOWN BUTTONS TO SET THE TIME AND SPEED***
- ***TIME IS IN MINUTES***
- ***SPEED IS __*100 (EXAMPLE: 15=1500RPM)***

SUMMARY SET UP STEPS

- UNPACKAGE CENTRIFUGE
- PLUG IN AND TURN ON
- OPEN LID AND REMOVE SHIPPING INSERT
- REMOVE AND SAVE TRANSPORT BOLTS AND RED SPACERS
- GREASE MOTOR SHAFT AND INSTALL ROTOR
- GREASE TRUNNIONS AND INSTALL BUCKETS
- BEGIN CENTRIFUGATION

COMMON ERRORS

- **ERROR CODE -1- (tachometer error)**
- **This is explained in the manual on page 28**

<u>Clearing procedure</u>

- ***Wait for rotor stand still (120 seconds after rotor stops)***
- ***Turn the power off and wait 10 seconds***
- ***Turn the power on***
- ***Check to make sure rotor is installed properly before spinning***

COMMON ERRORS

- **ERROR CODE -2- (mains interrupt)**
- **This is explained in the manual on page 28**

<u>Clearing procedure</u>

- ***Wait for rotor stand still***
- ***Open the lid***
- ***Press the start button***

COMMON ERRORS

- **ERROR CODE -d- (lid lock error)**
- **This is explained in the manual on page 28**

<u>Clearing procedure</u>

- ***Wait for rotor stand still***
- ***Turn off the power and wait 10 seconds***
- ***Turn on the power***

COMMON ERRORS

- **ERROR CODE -F- (tachometer error)**
- **This is explained in the manual on page 28**

<u>Clearing procedure</u>

- ***Wait for rotor stand still***
- ***Turn off power wait 10 seconds***
- ***Turn on power***
- ***Check to make sure rotor is installed properly before spinning***